ML fundamentals

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1 Introduction

Machine Learning Preliminaries introduce the core ideas required to understand modern OCR systems. We cover fundamental models such as linear and logistic regression, basic evaluation tools like train–test split and confusion matrix, and extend these ideas to neural networks, CNNs, and SVMs. Key concepts including activation functions, loss functions, hyperparameters, backpropagation, overfitting, and underfitting explain how models learn and generalize from data. In traditional OCR systems like PyTesseract, ML is mainly used in the classification stage after rule-based preprocessing, whereas modern OCR models apply these ML principles end-to-end.

2 Core idea

In simple terms, machine learning is a way to use data to learn patterns and make predictions without explicitly programming every rule. In today's world, large amounts of data are generated continuously, and manually analyzing this data is inefficient. A machine learning system addresses this by taking existing data, identifying meaningful relationships within it, and using those relationships to make decisions on new, unseen data.

The process works as follows: we start with data, choose a machine learning algorithm, and allow it to learn from the data. An algorithm defines how learning happens—how the system updates itself when it makes mistakes—while the machine learning model is the final learned mathematical representation produced after training. Learning happens by making predictions, measuring how wrong those predictions are using a loss function, and gradually improving the model using optimization techniques. Once trained, the model can generalize beyond the training data and produce useful outputs

3 Basic terminologies

- **Data / Dataset**- A collection of examples used for learning. Each example represents one observation (e.g., an image, a row in a table).
- **Feature (Input)**- Individual measurable properties of data used by the model (e.g., pixel values of an image).
- **Target / Label (Output)**- The value the model is expected to predict (e.g., character class in OCR).
- **Training Data**- The portion of data used to teach the model patterns.
- **Test Data**- Data not seen during training, used to evaluate how well the model generalizes.
- **Train–Test Split**- The process of dividing data into training and testing sets.
- **Algorithm**- The learning procedure or rule that updates the model using data (e.g., gradient descent).
- **Model**- The learned mathematical representation produced after training that makes predictions on new data.
- **Prediction**- The output produced by a trained model for a given input.

- **Loss Function**- A function that measures how wrong a model's prediction is compared to the true label.
- **Optimization**- The process of adjusting model parameters to minimize the loss.
- **Parameters**- Values learned by the model during training (e.g., weights and biases).
- **Hyperparameters**- User-defined settings that control learning behavior (e.g., learning rate, number of layers).
- **Epoch**- One complete pass of the training data through the model.
- **Overfitting**- When a model learns training data too well and performs poorly on new data.
- **Underfitting**- When a model is too simple to capture patterns in the data.
- **Confusion Matrix**- A table used to evaluate classification performance by comparing predictions with true labels.
- **Generalization**- The ability of a model to perform well on unseen data.
- **Supervised Learning**- A learning setup where the data comes with correct answers (labels). The model learns by comparing its predictions with these known outputs.
- **Unsupervised Learning**- A learning setup where no labels are provided. The model tries to discover patterns or structure in the data on its own.
- **Regression**- A type of problem where the target value is continuous (e.g., predicting a number).
- **Classification**- A type of problem where the target value is discrete or categorical (e.g., choosing one class out of many).

You can read more about train-test split here.

4 Linear Regression

Linear regression is a model that predicts a continuous value by fitting the best-possible straight line through the data.you can read more about it here.
Alternatively if you prefer videos you can find them here.
This playlist of videos by Andrew NG is also a good resource for Linear regression and to get a good intro into ML.

5 Logistic Regression

Logistic regression is a model that predicts which class an input belongs to by estimating probabilities and applying a decision boundary.
You can read more about it here.
Video resource here, the relevant topic is till video-7, multiclass classification.

6 Support Vector Machines (SVM)

Support Vector Machines is a model that separates data into classes by finding the best boundary that maximizes the margin between them.
Reading material here.
This video will give you an intuition of SVMs.

7 Overfitting Underfitting and Regularization

In practice, machine learning models can fail in two common ways: underfitting and overfitting. Underfitting occurs when the model is too simple to capture the underlying patterns in the data, while overfitting occurs when the model becomes too complex and starts memorizing the training data instead of learning general trends. These issues often become clear when we visualize training and test performance, such as by comparing errors or accuracy across datasets. To address overfitting, techniques like regularization are used, which limit model complexity by adding constraints or penalties during training. Regularization helps the model focus on important patterns, improving its ability to generalize and perform well on unseen data.
The problem of overfitting and regularization applied on linear and logistic regression is detailed in videos 8-11 of this playlist.
This article gives a good idea about regularization and its implementation in python.

8 Neural Networks

While traditional machine learning models work well for simpler patterns, they often struggle with complex, high-dimensional data such as images and text. Neural networks address this limitation by learning multiple levels of representation directly from data. Instead of relying on hand-crafted features, neural networks automatically learn useful features through stacked layers of computation. Each layer captures increasingly abstract patterns, allowing the model to handle non-linear relationships and complex structures. This makes neural networks especially powerful for vision-based tasks like OCR, where raw pixel information must be transformed into meaningful character and word representations.
You can read these medium articles to get a good introduction into neural networks

- article-1
- article-2

For videos you can refer to the first 4 videos of this playlist or till part-7 of this playlist.The second playlist contains more info so you are advised to skim through it

9 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a specialized type of neural network designed specifically for image and visual data. Unlike fully connected neural networks, which treat every input value equally, CNNs exploit the spatial structure of images by focusing on local regions through convolution operations. This allows CNNs to efficiently learn visual patterns such as edges, shapes, and textures while using fewer parameters. Because of this structure-aware learning,

CNNs perform significantly better on image-based tasks and are widely used in applications like image classification, object detection, and OCR, where understanding spatial relationships between pixels is critical.
you can gain a basic idea on the working of CNNs here.
This video also gives a good intuition on the working of CNNs.
This video by MIT gives you a deeper understanding and implementation steps.

Note

If you are facing difficulties with gradient descent you can refer to this video.